MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION/IMPROVEMENT/REAPPEARANCE EXAMINATION APRIL 2026

FOURTH SEMESTER

Discipline Specific Core Course [DSC]-MG4DSCPES200 –

SCIENCE OF SPORTS TRAINING

(2024 ADMISSION ONWARDS)

Answer Key

Duration: 1.5 Hours **Maximum Mark: 50**

ONE MARK QUESTION AND ANSWER

1 Power is defined as?

Ans: b. Work done per unit time

2 Which is a type of speed?

Ans: a. Flexibility speed

3 Periodization helps prevent

Ans :b . Overtraining and plateaus

4 For adolescents, priority should be given to

Ans. b.Long term athletic development

5 Exercise physiology helps in determining

Ans .d Training intensity and load

6 Ballistic stretching involves?

Ans.c. Bouncing movements

7 Plyometric intensity can be modified by

Ans.b. Box height, distance, and body position

8 Chatbots in sports coaching are mainly used for

Ans. b Athlete communication support

9. For beginners, recommended frequency is

Ans. b 2-3 days per week

10. **A gyroscope sensor in sports wearables is mainly used to measure?**

Ans.b. Rotational movement and body orientation

PART B

SHORT ANSWER 2 MARKS (answer any 5 questions)

11. What is training intensity?

Training intensity refers to the level of effort or load applied during a workout, often measured by factors like weight, speed, or physiological stress (e.g., % of maximum heart rate).

12. When to avoid static stretching?

Static stretching should be avoided before high-intensity or explosive activities, during acute injury, or when muscles are cold, as it may reduce performance or increase injury risk.

13. Give two agility drills.

1. Shuttle Runs: Sprint between two points, touching the line or cone each time.

2. Cone Weave: Weave through cones set in a zig-zag pattern.

3. Lateral Shuffles: Shuffle sideways between two points.

4. Box Drills: Sprint around a square or box-shaped course.

14 How does data analytics assist in injury prevention?

Data analytics helps identify patterns in workload, movement biomechanics, or recovery metrics, enabling tailored training adjustments and early risk detection to prevent injuries.

15 Explain how resistance training improves muscular strength based on scientific principles

1. Muscular Hypertrophy: Progressive overload causes micro-tears in muscle fibres, leading to repair and growth, increasing muscle size and strength.

2. Neural Adaptations: Enhanced motor unit recruitment and synchronization improve muscle activation, contributing to increased strength.

16 Sport specific considerations?

Ans 1. Specificity: Tailor training to mimic the sport's movements, intensity, and energy systems.

2. Injury Prevention: Address common injury patterns and incorporate exercises to strengthen vulnerable areas.

3. Movement pattern analysis

17 what is an electric lap counter

An electronic device used to track and display the number of laps completed by an athlete, typically in swimming or track events, providing accurate and efficient counting.

PART C (5 marks questions)

PART C .Short Essay (5 marks questions)

Answer any 4 Questions

18 .Describe the different types of training load and their significance in athletic preparation.

1. External load – measurable work done (e.g., weight lifted, distance run).

2. Internal load – physiological response (e.g., heart rate, lactate).

3. Acute load – short-term workload in a single session.

4. Chronic load – cumulative workload over weeks/months.

5. Significance: balancing external & internal loads optimizes performance and prevents overtraining.

19. Explain the strength-endurance continuum and its implications for training

1. Strength endurance

2. Muscular Endurance

3. Middle zone – mixed strength-endurance training for functional performance.

4. Implication: program design should place athletes on the continuum according to sport demands.

20. Explain the phases of the stretch-shortening cycle and their role in enhancing muscular power.

1. Eccentric phase

2. Amortization phase – brief transition to concentric action (must be minimized).

3. Concentric phase

4. Role: SSC improves power output in activities like jumps or sprints.

21 Describe the importance of hydration monitoring using biometric wearables.

1. Real-time sweat-rate data helps assess fluid loss.

2. Electrolyte balance tracking prevents dehydration or over-hydration.

3. Performance optimization by adjusting intake during training/competition.

4. Injury prevention by avoiding heat-related stress.

22. Describe how recovery science has advanced training practices.

1. Monitoring tools (HRV, sleep metrics) guide individualized recovery protocols.

2. Nutrition strategies (protein timing, anti-inflammatory foods) enhance repair.

3. Active recovery (low-intensity exercise, mobility work) improves blood flow.

4. Result: smarter periodization and reduced risk of overtraining.

23. Design a flexibility program for a gymnast addressing different stretching types.

1. Dynamic warm-up (leg swings, arm circles) for mobility before routines.

2. Static stretching (hold 20-30 s) for post-session muscle lengthening (hamstrings, shoulders).

3. PNF stretching (contract-relax) for targeted joint range improvement.

4. Flexibility maintenance – daily routine integrating all types to support gymnastics skills.

PART D (Essay 15 Marks)

Answer any 2 Questions

24 Explain the supercompensation theory in detail and its practical applications in periodized training programs.?

Supercompensation Theory explains how the body adapts to training stress. After a workout, performance temporarily decreases due to fatigue, then recovers, and finally improves beyond the original level.

Phases of Supercompensation

Initial Fatigue Phase

Training causes muscle damage and energy depletion

Performance decreases

Recovery Phase

Body restores energy stores (ATP, glycogen)

Repairs muscle tissue

Supercompensation Phase

Performance increases above baseline

Ideal time for next training session

Involution (Reversibility) Phase

If no training, gains are lost

Graph Explanation

Performance ↓ (fatigue) → returns to normal → ↑ (supercompensation peak) → ↓ if no training

Importance

Helps in planning training load

Prevents overtraining

Maximizes performance improvement

Application in Periodized Training

Periodization is the systematic planning of training.

1. Micro cycle (Weekly Plan)

Alternate hard and easy days

Allow recovery for supercompensation

2. Mesocycle (Monthly Plan)

Progressive overload followed by recovery weeks

3. Macrocycle (Yearly Plan)

Peak performance timed for competitions

Practical Example

Day 1: Heavy strength training

Day 2: Rest or light training

Day 3: Performance improves (supercompensation)

Conclusion

Supercompensation is essential for effective training. Proper timing of workouts leads to continuous improvement and peak performance.

25 . Discuss the factors affecting strength development and their consideration in personalized training.?

Strength is the ability to exert force. Its development depends on several physiological, psychological, and environmental factors.

Factors Affecting Strength Development

1. Muscle Size (Hypertrophy)

Larger muscles produce more force

2. Neural Factors

Motor unit recruitment

Coordination and firing rate

3. Age

Peak strength in early adulthood

Youth need lighter loads

4. Gender

Males generally have higher muscle mass

Females respond equally well to training

5. Nutrition

Protein intake essential for muscle growth

Energy balance important

6. Hormones

Testosterone, growth hormone influence strength

7. Training Type

Resistance training improves strength

Progressive overload is key

8. Rest and Recovery

Muscles grow during recovery

9. Genetics

Determines muscle fiber type (fast/slow twitch)

10. Motivation & Psychology

Effort and consistency affect outcomes

Considerations in Personalized Training

Assessment of Individual Fitness

Goal Setting (strength, endurance, rehab)

Customized Load & Intensity

Age-specific programming

Injury history consideration

Proper technique training

Recovery planning

Monitoring progress (tests, feedback)

Conclusion

Personalized training ensures safety, effectiveness, and optimal strength development for individuals.

26 .Design comprehensive plyometric programs for upper and lower body development.

Plyometric Training involves explosive movements using the stretch-shortening cycle to improve power.

Principles

Warm-up is essential

Quality over quantity

Adequate rest between sets

Progressive intensity

A. Lower Body Plyometric Program

Exercises

Squat jumps

Box jumps

Bounding

Depth jumps

Lateral hops

Program Design

Frequency: 2–3 times/week

Sets: 3–5

Reps: 8–12

Rest: 1–2 minutes

B. Upper Body Plyometric Program

Exercises

Medicine ball chest pass

Overhead throws

Plyometric push-ups

Side throws

Clap push-ups

Program Design

Frequency: 2–3 times/week

Sets: 3–4

Reps: 6–10

Rest: 1–2 minutes

Safety Considerations

Proper landing technique

Avoid hard surfaces

Suitable for trained individuals

Gradual progression

Benefits

Improves explosive strength

Enhances speed and agility

Better athletic performance

27 .How would you apply Beep test results to improve cardiovascular endurance in athletes?

Beep Test (also called shuttle run test) measures aerobic endurance and estimates VO_2 max.

Understanding Beep Test Results

Score = Level + Shuttle number

Higher level = better endurance

Used to assess fitness level

Application in Training

1. Fitness Assessment

Classify athletes (low, moderate, high fitness)

2. Training Zone Identification

Helps set intensity levels

3. Designing Endurance Program

Low score → basic aerobic training

High score → advanced interval training

4. Training Methods Based on Results

a. Continuous Training

Jogging, cycling for beginners

b. Interval Training

Work-rest cycles improve VO_2 max

c. High-Intensity Interval Training (HIIT)

Short bursts of high intensity

d. Sport-specific drills

Shuttle runs, agility drills

5. Monitoring Progress

Retest every 4–6 weeks

Compare scores to track improvement

6. Motivation Tool

Provides measurable goals

Example

Athlete scores Level 6 → Start moderate training

After 6 weeks → improves to Level 8